Characterizing graphs from diffused signals *

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1 Introduction

An important requirement in the field of signal processing on graphs is the need for a graph defining a domain on which observed signals evolve. When such a graph is available, one can compute the eigenbasis of the associated Laplacian matrix. Based on the spectral graph theory introduced by Chung in [1], it was shown by Shuman *et. al* [7] that this particular basis embeds a notion of frequency, where the smallest eigenvalues correspond to the lowest frequencies in classical Fourier analysis, and the biggest eigenvalues correspond to the highest frequencies. This important result has enabled researchers to develop numerous techniques to process signals defined on graphs. Exemples are convolution of signals [7] or wavelets on graphs [3], with applications to multiscale clustering [8] or transportation networks [4].

However, in the case when no graph is available, none of these techniques can be applied. Therefore, an important problem is how to recover a graph from a set of signals in order to process them efficiently.

In this presentation, we want to address this problem, by presenting extensions of our previous work in [5]. We consider the case when observed signals are issued from the diffusion of initially independent signals on an unknown graph structure, that we want to recover. We show that that the matrix representing this diffusion process is not unique, and that the space of admissible diffusion matrices forms a convex polytope. Then, we introduce one possible search strategy in this polytope to select a graph for which the main diagonal is null.

2 Summary of the presentation

We have previously shown in [5] that it is possible to recover a diffusion matrix from signals diffused on it, provided that there is a sufficient number of them. This work led to a result stating that the covariance matrix of the signals is equal to a power of the diffusion matrix $\mathbf{T} = \mathbf{D}^{-\frac{1}{2}}\mathbf{W}\mathbf{D}^{-\frac{1}{2}}$; where $\mathbf{W}$ is the adjacency matrix of the graph and $\mathbf{D}$ its diagonal matrix of degrees. A direct consequence is that the eigenvectors of the covariance matrix are the same as those of the diffusion matrix. We then proposed an optimization problem allowing us to recover the eigenvalues of the diffusion matrix.

However, this first work has some limiting assumptions, such as the number of signals being big enough, or the number of diffusion steps for all signals being fixed and known. In this presentation, we generalize our previous approach by removing these assumptions.

By construction, any diffusion matrix for a graph is a symetric positive matrix. If the eigenvectors of the diffusion matrix are known, for example using PCA [6] on the covariance matrix of the signals, then we can describe each of its cells as a linear combination of its eigenvalues. Therefore, the space of eigenvalues associated with acceptable diffusion matrices can be seen as a convex polytope, delimited by inequations enforcing the positivity of each cell.

If the original graph is simple, *i.e* if all elements in the diagonal of its adjacency matrix are null, then by construction it is also the case for its associated diffusion matrix. Therefore, the eigenvalues of such diffusion matrix can be found in the convex polytope by solving a simplex problem [2] in which we minimize the sum of the eigenvalues we want to recover. In Figure 1, we plot for a 3×3 simple graph the space of eigenvalues defining admissible diffusion matrices. The *bottom-left corner* of the polytope, which is found with our simplex method, minimizes the sum of the eigenvalues, and therefore corresponds to the set of eigenvalues minimizing the trace of the diffusion matrix.

When the number of signals is restricted, then the eigenvectors of the covariance matrix are not exactly equal to those of $\mathbf{T}$. This implies a deformation of the polytope, which tends to the *groundtruth* one as the number of signals increases. Figure 2 depicts the polytope associated with the same matrix as in Figure 1 when we use the eigenvectors obtained using PCA on 10 signals issued from a variable number of diffusions (in $[\![2; 5]\!]$) of independant signals.

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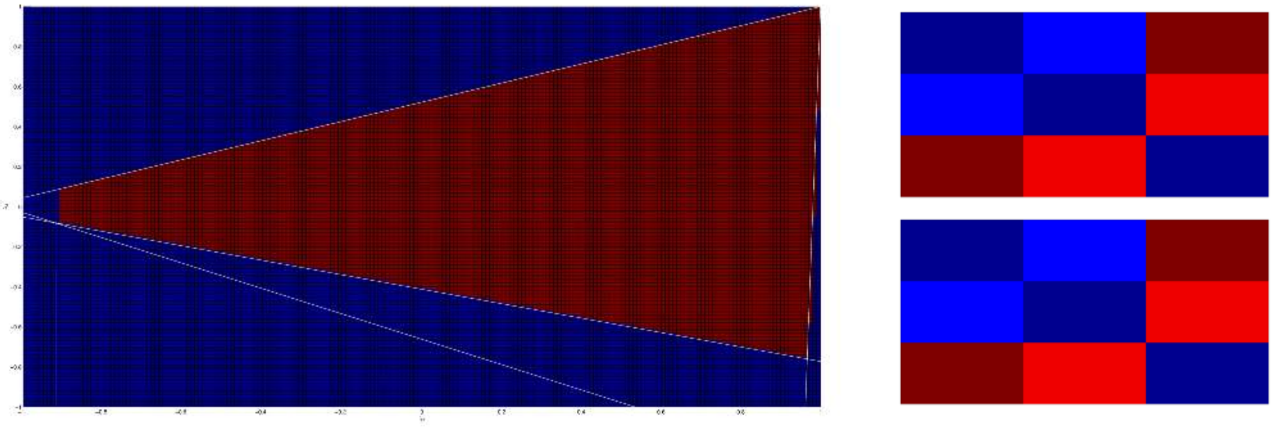

Figure 1: Convex polytope representing the eigenvalues leading to an admissible diffusion matrix, given the exact eigenvectors of $\mathbf{T}$. Here, λ_1 is set to 1 to impose a scale, while λ_2 and λ_3 are iterated over in steps of 0.01. The top-right matrix is the original diffusion matrix. The matrix on the bottom-right is the one recovered using our simplex method.

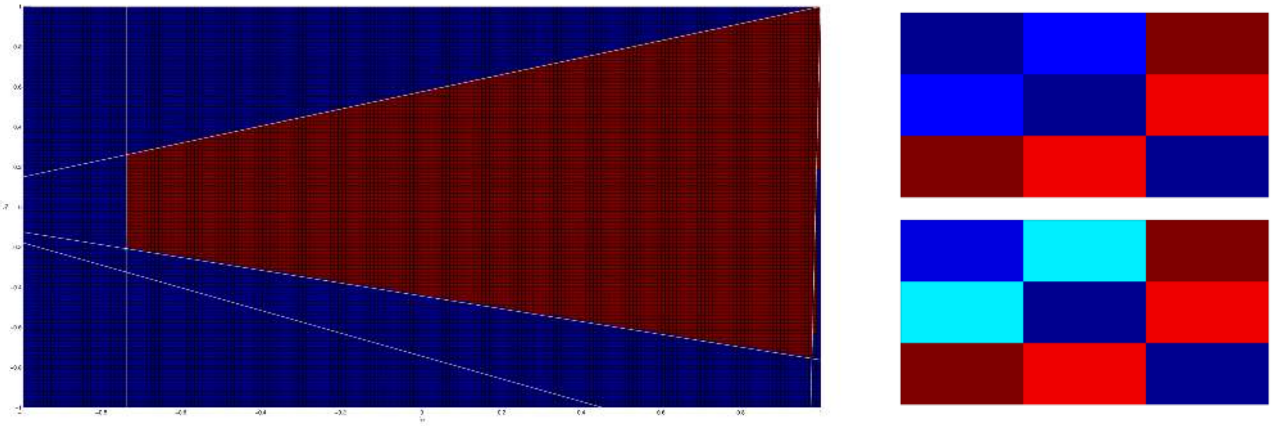

Figure 2: Convex polytope representing the eigenvalues leading to an admissible diffusion matrix, given the vectors obtained by PCA on 10 signals. Here, λ_1 is set to 1 to impose a scale, while λ_2 and λ_3 are iterated over in steps of 0.01. The top-right matrix is the original diffusion matrix. The matrix on the bottom-right is the one recovered using our simplex method.

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